

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} T_x & 0 & 0 \end{bmatrix}$$

$$T_x = T_{px}^{(1)}$$

* Rank 2 tensor (s).

$$\begin{bmatrix} 0 & 0 & T_{xz} \\ 0 & 0 & 0 \\ T_{xz} & 0 & 0 \end{bmatrix}$$

$$T_{xz} = T_{dxz}^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & 0 & T_{xz} \\ 0 & 0 & 0 \\ -T_{xz} & 0 & 0 \end{bmatrix}$$

$$T_{xz} = -M_{py}^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} T_{xxx} & 0 & 0 \\ T_{yyx} & 0 & 0 \\ T_{zzx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & T_{zzx} \\ 0 & T_{yyx} & 0 \end{bmatrix}$$

$$T_{xxx} = 2T_{fax}^{(1)} + 3T_{px}^{(1)}$$

$$T_{yyx} = -T_{fax}^{(1)} + T_{fbx}^{(1)} + T_{px}^{(1)}$$

$$T_{zzx} = -T_{fax}^{(1)} - T_{fbx}^{(1)} + T_{px}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ T_{yyx} & 0 & 0 \\ T_{zzx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -\frac{T_{zzx}}{2} \\ 0 & -\frac{T_{yyx}}{2} & 0 \end{bmatrix}$$

$$T_{yyx} = 2M_{dyz}^{(1)} + 2T_{px}^{(2)}$$

$$T_{zzx} = -2M_{dyz}^{(1)} + 2T_{px}^{(2)}$$

$$\frac{T_{zzx}}{2} = -M_{dyz}^{(1)} + T_{px}^{(2)}$$

$$\frac{T_{yyx}}{2} = M_{dyz}^{(1)} + T_{px}^{(2)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & T_{zzx} \\ 0 & T_{xyy} & 0 \end{bmatrix}$$

$$T_{zzz} = M_{dyz}^{(2)} + T_{px}^{(3)}$$

$$T_{xyy} = M_{dyz}^{(2)} - T_{px}^{(3)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & T_{xxx} & 0 \\ 0 & 0 & 0 & 0 & T_{yyz} & 0 \\ 0 & 0 & 0 & 0 & T_{zzz} & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{yzy} \\ T_{xxx} & T_{yyz} & T_{zzz} & 0 & 0 & 0 \\ 0 & 0 & 0 & T_{yzy} & 0 & 0 \end{bmatrix}$$

$$T_{xxx} = T_{dxz}^{(1)} + 2T_{dxz}^{(2)} - T_{gay}^{(1)} - T_{gby}^{(1)}$$

$$T_{yyz} = T_{dxz}^{(1)} + 2T_{gby}^{(1)}$$

$$T_{zzz} = T_{dxz}^{(1)} + 2T_{dxz}^{(2)} + T_{gay}^{(1)} - T_{gby}^{(1)}$$

$$T_{yzy} = T_{dxz}^{(2)} + 2T_{gby}^{(1)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & T_{xxx} & 0 \\ 0 & 0 & 0 & 0 & T_{yyz} & 0 \\ 0 & 0 & 0 & 0 & T_{zzz} & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{yzy} \\ -T_{xxx} & -T_{yyz} & -T_{zzz} & 0 & 0 & 0 \\ 0 & 0 & 0 & -T_{yzy} & 0 & 0 \end{bmatrix}$$

$$T_{xxx} = 2M_{fay}^{(1)} + 2M_{fby}^{(1)} - 2M_{py}^{(1)} + T_{dxz}^{(3)}$$

$$T_{yyz} = -4M_{fby}^{(1)} + T_{dxz}^{(3)}$$

$$T_{zzz} = -2M_{fay}^{(1)} + 2M_{fby}^{(1)} + 2M_{py}^{(1)} + T_{dxz}^{(3)}$$

$$T_{yzy} = 4M_{fay}^{(1)} + M_{py}^{(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} 0 & 0 & T_{yzy} \\ 0 & 0 & 0 \\ T_{yzy} & 0 & 0 \end{bmatrix}$$

$$T_{yzy} = 2T_{dxz}^{(4)}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & 0 & T_{yzy} \\ 0 & 0 & 0 \\ -T_{yzy} & 0 & 0 \end{bmatrix}$$

$$T_{yzy} = M_{py}^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & T_{xxx} & 0 \\ 0 & T_{yyz} & 0 \\ 0 & T_{zzz} & 0 \\ 0 & 0 & T_{yzy} \\ 0 & 0 & 0 \\ T_{yyz} & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
T_{xxzx} &= -M_{fay}^{(2)} - M_{fby}^{(2)} + M_{py}^{(3)} + 2T_{dxz}^{(5)} \\
T_{yyzx} &= 2M_{fay}^{(2)} + M_{py}^{(3)} + 2M_{py}^{(4)} \\
T_{zzzx} &= -M_{fay}^{(2)} + M_{fby}^{(2)} + M_{py}^{(3)} - 2T_{dxz}^{(5)} \\
T_{yzxy} &= -M_{fay}^{(2)} + M_{fby}^{(2)} + M_{py}^{(4)} + T_{dxz}^{(5)} \\
T_{xyyz} &= -M_{fay}^{(2)} - M_{fby}^{(2)} + M_{py}^{(4)} - T_{dxz}^{(5)}
\end{aligned}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & T_{yzxy} \\ T_{zzxx} & T_{zxyy} & T_{zzzz} & 0 & 0 & 0 \\ 0 & 0 & 0 & T_{xyyz} & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
T_{yzxy} &= -M_{fay}^{(3)} - M_{fby}^{(3)} + M_{py}^{(6)} - T_{dxz}^{(6)} \\
T_{zzxx} &= -M_{fay}^{(3)} - M_{fby}^{(3)} + M_{py}^{(5)} + 2T_{dxz}^{(6)} \\
T_{zxyy} &= 2M_{fay}^{(3)} + M_{py}^{(5)} + 2M_{py}^{(6)} \\
T_{zzzz} &= -M_{fay}^{(3)} + M_{fby}^{(3)} + M_{py}^{(5)} - 2T_{dxz}^{(6)} \\
T_{xyyz} &= -M_{fay}^{(3)} + M_{fby}^{(3)} + M_{py}^{(6)} + T_{dxz}^{(6)}
\end{aligned}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & T_{xxzx} & 0 & 0 & T_{xxzx} & 0 \\ 0 & 0 & 0 & 0 & T_{yyzx} & 0 & 0 & T_{yyzx} & 0 \\ 0 & 0 & 0 & 0 & T_{zzzx} & 0 & 0 & T_{zzzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{yzxy} & 0 & 0 & T_{yzxy} \\ T_{zzxx} & T_{zxyy} & T_{zzzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & T_{xyyz} & 0 & 0 & T_{xyyz} & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
T_{xxzx} &= 2M_{fay}^{(1)} - M_{fay}^{(2)} + 2M_{fby}^{(1)} - M_{fby}^{(2)} - 2M_{py}^{(1)} + M_{py}^{(3)} + T_{dxz}^{(1)} + 2T_{dxz}^{(2)} + T_{dxz}^{(3)} + 2T_{dxz}^{(5)} - T_{gay}^{(1)} - T_{gby}^{(1)} \\
T_{xxzx} &= 2M_{fay}^{(1)} + M_{fay}^{(2)} + 2M_{fby}^{(1)} + M_{fby}^{(2)} - 2M_{py}^{(1)} - M_{py}^{(3)} + T_{dxz}^{(1)} + 2T_{dxz}^{(2)} + T_{dxz}^{(3)} - 2T_{dxz}^{(5)} - T_{gay}^{(1)} - T_{gby}^{(1)} \\
T_{yyzx} &= 2M_{fay}^{(2)} - 4M_{fby}^{(1)} + M_{py}^{(3)} + 2M_{py}^{(4)} + T_{dxz}^{(1)} + T_{dxz}^{(3)} + 2T_{gby}^{(1)} \\
T_{yyzx} &= -2M_{fay}^{(2)} - 4M_{fby}^{(1)} - M_{py}^{(3)} - 2M_{py}^{(4)} + T_{dxz}^{(1)} + T_{dxz}^{(3)} + 2T_{gby}^{(1)} \\
T_{zzzx} &= -2M_{fay}^{(1)} - M_{fay}^{(2)} + 2M_{fby}^{(1)} + M_{fby}^{(2)} + 2M_{py}^{(1)} + M_{py}^{(3)} + T_{dxz}^{(1)} + 2T_{dxz}^{(2)} + T_{dxz}^{(3)} - 2T_{dxz}^{(5)} + T_{gay}^{(1)} - T_{gby}^{(1)} \\
T_{zzzx} &= -2M_{fay}^{(1)} + M_{fay}^{(2)} + 2M_{fby}^{(1)} - M_{fby}^{(2)} + 2M_{py}^{(1)} - M_{py}^{(3)} + T_{dxz}^{(1)} + 2T_{dxz}^{(2)} + T_{dxz}^{(3)} + 2T_{dxz}^{(5)} + T_{gay}^{(1)} - T_{gby}^{(1)} \\
T_{yzxy} &= 4M_{fay}^{(1)} - M_{fay}^{(2)} + M_{fby}^{(2)} + M_{py}^{(1)} + M_{py}^{(4)} + T_{dxz}^{(2)} + T_{dxz}^{(5)} + 2T_{gby}^{(1)} \\
T_{yzxy} &= 4M_{fay}^{(1)} + M_{fay}^{(2)} - M_{fby}^{(2)} + M_{py}^{(1)} - M_{py}^{(4)} + T_{dxz}^{(2)} - T_{dxz}^{(5)} + 2T_{gby}^{(1)} \\
T_{zzxx} &= -2M_{fay}^{(1)} - 2M_{fby}^{(1)} + 2M_{py}^{(1)} + T_{dxz}^{(1)} + 2T_{dxz}^{(2)} - T_{dxz}^{(3)} - T_{gay}^{(1)} - T_{gby}^{(1)} \\
T_{zxyy} &= 4M_{fby}^{(1)} + T_{dxz}^{(1)} - T_{dxz}^{(3)} + 2T_{gby}^{(1)} \\
T_{zzzz} &= 2M_{fay}^{(1)} - 2M_{fby}^{(1)} - 2M_{py}^{(1)} + T_{dxz}^{(1)} + 2T_{dxz}^{(2)} - T_{dxz}^{(3)} + T_{gay}^{(1)} - T_{gby}^{(1)} \\
T_{xyyz} &= -4M_{fay}^{(1)} - M_{fay}^{(2)} - M_{fby}^{(2)} - M_{py}^{(1)} + M_{py}^{(4)} + T_{dxz}^{(2)} - T_{dxz}^{(5)} + 2T_{gby}^{(1)} \\
T_{xyyz} &= -4M_{fay}^{(1)} + M_{fay}^{(2)} + M_{fby}^{(2)} - M_{py}^{(1)} - M_{py}^{(4)} + T_{dxz}^{(2)} + T_{dxz}^{(5)} + 2T_{gby}^{(1)}
\end{aligned}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & T_{yzxy} & 0 & 0 & T_{yzxy} \\ T_{zzxx} & T_{zxyy} & T_{zzzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & T_{xyyz} & 0 & 0 & T_{xyyz} & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
T_{yzxy} &= -M_{fay}^{(3)} - M_{fby}^{(3)} + M_{py}^{(2)} + M_{py}^{(6)} + 2T_{dxz}^{(4)} - T_{dxz}^{(6)} \\
T_{yzyx} &= -M_{fay}^{(3)} - M_{fby}^{(3)} - M_{py}^{(2)} + M_{py}^{(6)} - 2T_{dxz}^{(4)} - T_{dxz}^{(6)} \\
T_{zxzx} &= -M_{fay}^{(3)} - M_{fby}^{(3)} + M_{py}^{(5)} + 2T_{dxz}^{(6)} \\
T_{zxyy} &= 2M_{fay}^{(3)} + M_{py}^{(5)} + 2M_{py}^{(6)} \\
T_{zxxz} &= -M_{fay}^{(3)} + M_{fby}^{(3)} + M_{py}^{(5)} - 2T_{dxz}^{(6)} \\
T_{xyyz} &= -M_{fay}^{(3)} + M_{fby}^{(3)} - M_{py}^{(2)} + M_{py}^{(6)} + 2T_{dxz}^{(4)} + T_{dxz}^{(6)} \\
T_{xyzy} &= -M_{fay}^{(3)} + M_{fby}^{(3)} + M_{py}^{(2)} + M_{py}^{(6)} - 2T_{dxz}^{(4)} + T_{dxz}^{(6)}
\end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix}
0 & 0 & T_{xxxx} \\
0 & 0 & 0 \\
T_{zzzx} & 0 & 0 \\
T_{yyzx} & 0 & 0 \\
0 & 0 & T_{zzxz} \\
0 & 0 & 0 \\
0 & 0 & 0 \\
T_{zzxx} & 0 & 0 \\
0 & 0 & T_{xyyz} \\
0 & T_{xyyz} + \frac{T_{zzxz}}{2} - \frac{T_{zzzx}}{2} & 0
\end{bmatrix}$$

$$\begin{aligned}
T_{xxxx} &= T_{dxz}^{(1)} + 2T_{dxz}^{(2)} + T_{dxz}^{(3)} - 2T_{dxz}^{(5)} - T_{gay}^{(1)} - T_{gby}^{(1)} \\
T_{zzzx} &= T_{dxz}^{(1)} + 2T_{dxz}^{(2)} + T_{dxz}^{(3)} - 2T_{dxz}^{(5)} + T_{gay}^{(1)} - T_{gby}^{(1)} \\
T_{yyzx} &= T_{dxz}^{(1)} + T_{dxz}^{(3)} + 2T_{gby}^{(1)} \\
T_{zzxz} &= T_{dxz}^{(1)} + 2T_{dxz}^{(2)} + T_{dxz}^{(3)} + 2T_{dxz}^{(5)} + T_{gay}^{(1)} - T_{gby}^{(1)} \\
T_{zzxx} &= T_{dxz}^{(1)} + 2T_{dxz}^{(2)} - T_{dxz}^{(3)} - T_{gay}^{(1)} - T_{gby}^{(1)} \\
T_{xyyz} &= T_{dxz}^{(2)} - T_{dxz}^{(5)} + 2T_{gby}^{(1)}
\end{aligned}$$